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Enhancement of multiple monitor display capabilities

Abstract

An information handling system includes first and second graphics processing units, and a processor. The first graphics processing unit includes first, second, and third display ports. The second graphics processing unit includes fourth, fifth, and sixth display ports. The processor receive a buffer request that identifies a number of frame buffers to be created. The processor call a graphics multiple buffer render create buffer command to divide a graphics multiple buffer render buffer into multiple buffers. The multiple buffers of the graphics multiple buffer render buffer are based on the number of frame buffers identified in the buffer request. The processor create multiple frame buffers. The number of the frame buffers is the same as a number of the buffers of the graphics multiple buffer render buffer. Each of the frame buffers provide data to a different one of the first, second, third, fourth, fifth, and sixth display ports.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to information handling systems, and more particularly relates to an enhancement of multiple monitor display capabilities.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications

among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

(3) An information handling system includes first and second graphics processing units, and a processor. The first graphics processing unit includes first, second, and third display ports. The second graphics processing unit includes fourth, fifth, and sixth display ports. The processor may communicate with both the first graphics processing unit and with the second graphics processing unit. The processor may receive a buffer request that identifies a number of frame buffers to be created. The processor may call a graphics multiple buffer render create buffer command to divide a graphics multiple buffer render buffer into multiple buffers. The multiple buffers of the graphics multiple buffer render buffer are based on the number of frame buffers identified in the buffer request. The processor may create multiple frame buffers. The number of the frame buffers is the same as a number of the buffers of the graphics multiple buffer render buffer. Each of the frame buffers provide data to a different one of the first, second, third, fourth, fifth, and sixth display ports.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:
- (2) FIG. 1 is a block diagram of a system including an information handling system and multiple monitors according to at least one embodiment of the present disclosure;
- (3) FIG. 2 is a block diagram of a software stack and operating system for an information handling system according to at least one embodiment of the present disclosure;
- (4) FIG. 3 is a flow diagram of a method for creating multiple frame buffers for high resolution display on multiple display devices according to at least one embodiment of the present disclosure;
- (5) FIG. 4 is a flow diagram of a method for creating multiple frame buffers in an information handling system according to at least one embodiment of the present disclosure; and
- (6) FIG. 5 is a block diagram of a general information handling system according to an embodiment of the present disclosure.

(7) The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(8) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

(9) FIG. 1 illustrates a system **100** including an information handling system **102** and multiple monitors or display devices **104, 106, 108, 110, 112, and 114** (display devices **104-114**) according to at least one embodiment of the present disclosure. For purposes of this disclosure, an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer (such as a desktop or laptop), tablet computer, mobile device (such as a personal digital assistant (PDA) or smart phone), server (such as a blade server or rack server), a network storage device, or any other suitable device

and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, touchscreen and/or a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

(10) Information handling system **102** is in physical communication with display devices **104-114**. Information handling system **102** includes a processor **120** and graphics processing units (GPUs) **122** and **124**. Processor **120** includes multiple components including, but not limited to, a software stack **130** and an operating system (OS) **132**. In an example, code associated with software stack **130** and OS **132** may be stored within a memory internal to processor **120** or in a memory external to the processor, such as memory **520** or **525** of FIG. 5. In certain examples, GPUs **122** and **124** may have any suitable number of display ports and each display port may communicate with and provide rendering data to a different one of display devices **104-114**. For example, GPU **122** may include one display port to communicate with display device **104**, another display port to communicate with display device **106**, and even another display port to communicate with display device **108**. Similarly, GPU **124** may include one display port to communicate with display device **110**, another display port to communicate with display device **112**, and even another display port to communicate with display device **114**. Information handling system **102** may include any suitable number of additional components without varying from the scope of this disclosure.

(11) In previous information handling systems, the size of GPU memories may limit a maximum window size created by applications. In these information handling systems, the memory limitation may be represented as a maximum texture size, such as 16384 (2.sup.14) pixels. Based on this maximum texture size, the maximum buffer allocation may be 1 GB. A display device with 4k resolution has a pixel width of 3840. Based on the pixel width of 4k display devices, a maximum texture size of 16384 may only support four complete 4k display devices. Information handling system **102** may be improved by increasing buffer memory capacity to support more than four 4k display devices. Information handling system **100** may be further improved by processor **120** leveraging a maximum number of display ports that GPUs **122** and **124** may support. Also, processor **120** may dynamically detect and utilize the maximum number of display ports that GPUs **122** and **124** may support. Components of stack **130** and OS **132** will be described with respect to FIG. 2.

(12) FIG. 2 illustrates software stack **130** and OS **132** according to at least one embodiment of the present disclosure. Stack **130** includes an application/client layer **210**, a display server layer **212**, a graphics dispatcher layer **214**, a graphic multiple buffer render (GMBR) layer **216**, and a GPU specific layer **218**. OS **132** includes a device manager **220** and a frame buffer **222**. Stack **130** and OS **132** may both include any suitable number of additional components without varying from the scope of this disclosure.

(13) In an example, processor **120** of FIG. 1 may perform any of the operations described with respect to components of stack **130**, such as application/client layer **210**, display server layer **212**, graphics dispatcher layer **214**, graphic multiple buffer render (GMBR) layer **216**, and GPU specific layer **218**. Additionally, processor **120** of FIG. 1 may perform any of the operations described with respect to components of OS **132** including, but not limited to, device manager **220** and frame buffer **222**. Operations described with respect to application/client layer **210**, display server layer **212**, graphics dispatcher layer **214**, graphic multiple buffer render (GMBR) layer **216**, GPU specific layer **218**, device manager **220**, and frame buffer **222** may be understood to be performed by processor **120**, GPU **122**, or GPU **124**. The operations of components within information handling system **102** will be described with respect to FIGS. 1 and 2.

(14) During an initialization of information handling system **102**, GMBR layer **216** may detect details associated with display devices **104-114**. In an example, the details associated with display devices **104-114** may include, but are not limited to, the number of display devices and a resolution for each of the display devices. In an example, processor **120**, via application/client layer **210**, may request an OS buffer to be created. In certain examples, application layer **210** provide the buffer creation request to graphics dispatcher layer **214** in any suitable manner. For example, application layer **210** may provide the buffer creation request directly to graphics dispatcher layer **214** or may provide the request to the graphics dispatcher layer via display server layer **212**. In certain examples, the buffer creation request or call may include data identifying a number of buffers to create. The number of buffers may be based on a number of display devices **104-114** to have data provided to and the resolution of the display displays. For example, the create buffer call may indicate any suitable number of display devices, such as six, seven, eight, nine, or the like, and may indicate that the resolution for each of the display devices is a 4k resolution. In an example, components, such as OS **204** and graphics dispatcher layer **214**, may already have data identifying GMBR layer **216**.

(15) In an example, graphics dispatcher layer **214** may provide a create buffer call to GMBR layer **216**. In response to the create buffer call, GMBR layer **216** may provide a create buffer call to GPUs **122** and **124**. In an example, GMBR layer **216** may determine a number of buffers to create based on the GMBR create buffer call relayed through graphics dispatcher layer **214**. At operation **332**, GMBR layer **216** may utilize a create buffer loop to create the number of buffers identified in the GMBR create buffer call. In certain examples, GMBR layer **216** may divide a GMBR buffer **224** into multiple subset buffers, such as buffers **230**, **232**, **234**, **236**, **238**, and **240**. In an example, the number of subset buffers created within GMBR buffer **224** may be equal to the number of display devices. For example, information handling system **102** is in communication with six display devices **104-114** and GMBR layer **216** may create six buffers **230**, **232**, **234**, **236**, **238**, and **240** within GMBR buffer **224**. In an example, GMBR layer **216** may utilize a loop operation to create the number of buffers by dividing a single GMBR buffer **224** into multiple smaller buffers **230**, **232**, **234**, **236**, **238**, and **240**.

(16) After the multiple GMBR buffers **230**, **232**, **234**, **236**, **238**, and **240** have been created, GMBR layer **216** returns a GMBR logic buffer object to application/client layer **210**. During application/client layer **210** may provide a render API call to graphics dispatcher layer **214**. Based on the render API call, graphics dispatcher layer **214** provides a GMBR function call to GMBR layer **216**. In an example, GMBR layer **216** may determine whether a buffer management logic should be performed based on a number of display devices connected to information handling system **102**. If the number of display devices **104-114** is less than a predetermined number of devices, such as four display devices, the buffer management logic does not need to be performed. Based on buffer management logic not needing to be performed, GMBR layer **216** may call GPU specific layer **218** to create a corresponding frame buffer, such as frame buffer **222**.

(17) If the number of display devices **104-114** is more than the predetermined number of devices, the buffer management logic does need to be performed. In an example, the GMBR layer **216** may identify the number of frame buffers to be created for GPUs **122** and **124**. GMBR layer **216** may utilize a create buffer loop to create the number of frame buffers, such as frame buffers **250**, **252**, **254**, **256**, **258**, and **260**, identified in the create frame buffer call. In an example, the frame buffers may be created by dividing frame buffer **222** into multiple smaller frame buffers **250**, **252**, **254**, **256**, **258**, and **260**. In certain examples, frame buffers **250**, **252**, and **254** may be associated with GPU **122**, and frame buffers **256**, **258**, and **260** may be associated with GPU **124**.

(18) Based on the multiple buffers **250**, **252**, **254**, **256**, **258**, and **260**, each of GPUs **122** and **124** may render images on three different display devices and rendered images may have 4k resolution. In an example, GPU **122** may render data from buffer **250** as an image on display device **104**, may render data from buffer **252** as an image on display device **106**, and may render data from buffer

254 as an image on display device 108. Similarly, GPU 124 may render data from buffer 256 as an image on display device 110, may render data from buffer 258 as an image on display device 112, and may render data from buffer 260 as an image on display device 114.

(19) FIG. 3 is a flow diagram of a method 300 for creating multiple frame buffers for high resolution display on multiple display devices according to at least one embodiment of the present disclosure. The operations of method 300 are performed by an application 302, an operating system (OS) 304, a dispatcher 306, a graphics multiple buffer render (GMBR) 308, and a graphics processing unit backend 310. In an example, application 302, OS 304, dispatcher 306, and GMBR 308 may be executed by a processor of an information handling system. It will be readily appreciated that not every operation set forth in this flow diagram is always necessary, and that certain operations of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure.

(20) At operation 320, OS 304 detects GMBR 308. In response to the detection of GMBR 308, OS 304 may provide a request to synchronize monitor or display device layout information at operation 322. At operation 324, application 302 may provide a window buffer request to OS 304. In an example, the buffer request may be associated with a render buffer to provide data to one or more display devices, such as display devices 104-114 of FIG. 1, for providing image on the display screens. In certain examples, operations by application 302 may not change after the buffer request, and OS 304 may own or control operations of the render buffer.

(21) At operation 326, OS 304 may provide a create buffer call to dispatcher 306. In certain examples, the create buffer call may include data identifying a number of buffers to create. The number of buffers may be based on a number of display devices to have data provided to and the resolution of the display displays. For example, the create buffer call may indicate any suitable number of display devices, such as six, seven, eight, nine, or the like, and may indicate that the resolution for each of the display devices is a 4k resolution. In an example, components, such as OS 304 and dispatcher 306, may already have data identifying GMBR 308.

(22) Based on the prior identification of GMBR 308, dispatcher 306 may provide a create buffer call to GMBR 308 at operation 328. In response to the create buffer call, GMBR 308 may provide a create buffer call to GPU backend 310 at operation 330. In an example, GMBR 308 may determine a number of buffers to create based on the GMBR create buffer call relayed from OS 304 through dispatcher 306. At operation 332, GMBR 308 may utilize a create buffer loop to create the number of buffers identified in the GMBR create buffer call. In certain examples, the number of buffers created may be substantially equal to the number of display devices, such as display devices 104-114 of FIG. 1, in communication with the information handling system. In an example, multiple operations 330 within loop operation 332 may create the number of buffers by dividing a single GMBR buffer into multiple smaller buffers. In certain examples, GMBR 308 and the multiple GMBR buffers may be part of a software stack for a processor of the information handling system.

(23) After the multiple GMBR buffers have been created, GMBR 308 returns a GMBR logic buffer object at operation 334. In an example, the GMBR logic buffer object may include any suitable data associated with the multiple GMBR buffers created during the multiple iterations of operation 330. At operation 336, OS 304 returns or provides the identities of the create GMBR buffers to application 302. At operation 338, application 302 provides a render API call to dispatcher 306. Based on the render API call, dispatcher 306 provides a GMBR function call to GMBR 308 at operation 340. In an example, the GMBR function may identify the number of frame buffers to be created for one or more GPUs. At operation 342, GMBR 308 creates a frame buffer. At operation 344, GMBR 308 may utilize a create buffer loop to create the number of frame buffers identified in the create frame buffer call. In an example, the frame buffers may be created by dividing a single large frame buffer into multiple smaller frame buffers.

(24) At operation 346, application 302 performs a swap buffer action, and provide the swap buffer action to dispatcher 306. In an example, the swap buffer action may be performed at any suitable

time including, but not limited to, after a first one of the GMBR buffers has rendered the data to the corresponding frame buffer, and additional data is to be written to another GMBR buffer. At operation **348**, dispatcher **306** provides the swap buffer request to GMBR **308**, which in turn may provide the data to a frame buffer corresponding to a currently selected GMBR buffer. At block **350**, GMBR **308** notifies OS **304** that the current GMBR buffer is committed to a particular frame buffer. At operation **352**, OS **304** renders the data from the GMBR buffer of the software stack to a corresponding frame buffer. At block **354**, the frame buffer provides the data to a display device or monitor. In an example, the data may be provided to the display device via a device manager.

(25) FIG. **4** illustrates a flow of a method **400** for creating multiple frame buffers in an information handling system according to at least one embodiment of the present disclosure, starting at block **402**. In an example, method **400** may be performed by any suitable component including, but not limited to, processor **120** of FIG. **1**. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary, and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure.

(26) At block **404**, a number of display devices in communication with an information handling system is determined. In an example, each display device may be in communication with the information handling system via a different display port. The number of display ports in each GPU of the information handling system may be determined or detected. In certain examples, a first group of the display ports, such as three display ports, are within a first GPU and a second group of the display ports, such as three display ports, are within a second GPU. In an example, a resolution of the output of each of the display ports is determined, such as a 4k resolution.

(27) At block **406**, a GMBR buffer is divided into multiple subset buffers. In an example, a number of the subset of buffers in the GMBR buffer is based on the number of display devices in communication with the information handling system. At block **408** a render application programming interface (API) call to request a buffer object is received. Based on the render API call, a buffer position and size for a frame buffer is calculated.

(28) At block **410** multiple frame buffers are created based on the render API call. The frame buffers are physical buffers within the information handling system. In an example, the frame buffers may be created by dividing a single large frame buffer into multiple smaller frame buffers. In certain examples, a number of the frame buffers created may be the same as the number of the buffers of the GMBR buffer. At block **412**, each of the multiple frame buffers is associated with a different display port of the GPUs of the information handling system, and the flow ends at block **414**. Each of the frame buffers may provide data to a different one of display ports in the GPUs of the information handling system.

(29) FIG. **5** shows a generalized embodiment of an information handling system **500** according to an embodiment of the present disclosure. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **500** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **500** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **500** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **500** can include one or more storage devices that can store machine-

executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **500** can also include one or more buses operable to transmit information between the various hardware components.

(30) Information handling system **500** can include devices or modules that embody one or more of the devices or modules described below and operates to perform one or more of the methods described below. Information handling system **500** includes a processors **502** and **504**, an input/output (I/O) interface **510**, memories **520** and **525**, a graphics interface **530**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **540**, a disk controller **550**, a hard disk drive (HDD) **554**, an optical disk drive (ODD) **556**, a disk emulator **560** connected to an external solid state drive (SSD) **562**, an I/O bridge **570**, one or more add-on resources **574**, a trusted platform module (TPM) **576**, a network interface **580**, a management device **590**, and a power supply **595**. Processors **502** and **504**, I/O interface **510**, memory **520**, graphics interface **530**, BIOS/UEFI module **540**, disk controller **550**, HDD **554**, ODD **556**, disk emulator **560**, SSD **562**, I/O bridge **570**, add-on resources **574**, TPM **576**, and network interface **580** operate together to provide a host environment of information handling system **500** that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system **500**.

(31) In the host environment, processor **502** is connected to I/O interface **510** via processor interface **506**, and processor **504** is connected to the I/O interface via processor interface **508**. Memory **520** is connected to processor **502** via a memory interface **522**. Memory **525** is connected to processor **504** via a memory interface **527**. Graphics interface **530** is connected to I/O interface **510** via a graphics interface **532** and provides a video display output **536** to a video display **534**. In a particular embodiment, information handling system **500** includes separate memories that are dedicated to each of processors **502** and **504** via separate memory interfaces. An example of memories **520** and **530** include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof.

(32) BIOS/UEFI module **540**, disk controller **550**, and I/O bridge **570** are connected to I/O interface **510** via an I/O channel **512**. An example of I/O channel **512** includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface **510** can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I.sup.2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module **540** includes BIOS/UEFI code operable to detect resources within information handling system **500**, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module **540** includes code that operates to detect resources within information handling system **500**, to provide drivers for the resources, to initialize the resources, and to access the resources.

(33) Disk controller **550** includes a disk interface **552** that connects the disk controller to HDD **554**, to ODD **556**, and to disk emulator **560**. An example of disk interface **552** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **560** permits SSD **564** to be connected to information handling system **500** via an external interface **562**. An example of external interface **562** includes a USB interface, an IEEE **4394** (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive **564** can be disposed within

information handling system **500**.

(34) I/O bridge **570** includes a peripheral interface **572** that connects the I/O bridge to add-on resource **574**, to TPM **576**, and to network interface **580**. Peripheral interface **572** can be the same type of interface as I/O channel **512** or can be a different type of interface. As such, I/O bridge **570** extends the capacity of I/O channel **512** when peripheral interface **572** and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel **572** when they are of a different type. Add-on resource **574** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **574** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **500**, a device that is external to the information handling system, or a combination thereof.

(35) Network interface **580** represents a NIC disposed within information handling system **500**, on a main circuit board of the information handling system, integrated onto another component such as I/O interface **510**, in another suitable location, or a combination thereof. Network interface device **580** includes network channels **582** and **584** that provide interfaces to devices that are external to information handling system **500**. In a particular embodiment, network channels **582** and **584** are of a different type than peripheral channel **572** and network interface **580** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **582** and **584** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **582** and **584** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

(36) Management device **590** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device (CPLD), and the like, which operate together to provide the management environment for information handling system **500**. In particular, management device **590** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I2C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **500**, such as system cooling fans and power supplies. Management device **590** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **500**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **500**.

(37) Management device **590** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **500** when the information handling system is otherwise shut down. An example of management device **590** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like. Management device **590** may further include associated memory devices, logic devices, security devices, or the like, as needed, or desired.

(38) Although only a few exemplary embodiments have been described in detail herein, those

skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.

Claims

1. An information handling system comprising: a first graphics processing unit including first, second, and third display ports; a second graphics processing unit including fourth, fifth, and sixth display ports; and a processor to communicate with both the first graphics processing unit and with the second graphics processing unit, the processor to: receive a buffer request, wherein the buffer request identifies a number of frame buffers to be created; call a graphics multiple buffer render create buffer command to divide a graphics multiple buffer render buffer into a plurality of buffers, wherein the buffers of the graphics multiple buffer render buffer are based on the number of frame buffers identified in the buffer request; determine a number of display devices connected to the information handling system; in response to the number of display devices being below a threshold number of display devices, create a plurality of frame buffers, wherein a number of the frame buffers is the same as a number of the buffers of the graphics multiple buffer render buffer, wherein each of the frame buffers provides data to a different one of the first, second, third, fourth, fifth, and sixth display ports; and in response to the number of display devices being above threshold number of display devices, perform a buffer management logic prior to the plurality of frame buffers being created.
2. The information handling system of claim 1, wherein the processor to: wherein the number of the frame buffers is the same as the number of display devices.
3. The information handling system of claim 1, wherein the reception of the buffer request includes the processor to: receive a render application programming interface call to request a buffer object.
4. The information handling system of claim 1, wherein the processor further to: calculate a buffer position and size; and create the number of the buffers of the graphics multiple buffer render buffer.
5. The information handling system of claim 1, wherein the processor further to: detect an output resolution for each of the of the first, second, third, fourth, fifth, and sixth display ports.
6. The information handling system of claim 1, wherein the processor further to: detect a number of display ports in both the first and second graphics processing units.
7. The information handling system of claim 1, wherein the frame buffers are physical buffers within the information handling system.
8. The information handling system of claim 1, wherein each of the first, second, third, fourth, fifth, and sixth display ports outputs 4k resolution data.
9. A method comprising: receiving, by a processor of an information handling system, a buffer request, wherein the buffer request identifies a number of frame buffers to be created; calling, by the processor, a graphics multiple buffer render create buffer command to divide a graphics multiple buffer render buffer into a plurality of buffers, wherein the buffers of the graphics multiple buffer render buffer are based on the number of frame buffers identified in the buffer request; determine a number of display devices connected to the information handling system; in response to the number of display devices being below a threshold number of display devices, creating a plurality of frame buffers, wherein a number of the frame buffers is the same as a number of the buffers of the graphics multiple buffer render buffer, wherein each of the frame buffers provides data to a different one of first, second, third, fourth, fifth, and sixth display ports in first and second graphics processing units of the information handling system; and in response to the number of

display devices being above threshold number of display devices, performing a buffer management logic prior to the plurality of frame buffers being created.

10. The method of claim 9, further comprising: wherein the number of the frame buffers is the same as the number of display devices.

11. The method of claim 9, wherein the receiving of the buffer request, includes the method further comprising: receiving a render application programming interface call to request a buffer object.

12. The method of claim 9, further comprising: calculating a buffer position and size; and creating the number of the buffers of the graphics multiple buffer render buffer.

13. The method of claim 9, further comprising: detecting an output resolution for each of the of the first, second, third, fourth, fifth, and sixth display ports.

14. The method of claim 9, further comprising: detecting a number of display ports in both the first and second graphics processing units.

15. The method of claim 9, wherein the frame buffers are physical buffers within the information handling system.

16. The method of claim 9, wherein each of the first, second, third, fourth, fifth, and sixth display ports outputs 4k resolution data.

17. An information handling system comprising: a first graphics processing unit including first, second, and third display ports; a second graphics processing unit including fourth, fifth, and sixth display ports; and a processor to communicate with both the first graphics processing unit and with the second graphics processing unit, the processor to: determine a number of display devices in communication with the first and second graphics processing units; receive a buffer request, wherein the buffer request identifies a number of frame buffers to be created, wherein the number of the frame buffers to be created is the same as the number of display devices; calculate a buffer position and size of a graphics multiple buffer render buffer; call a graphics multiple buffer render create buffer command to divide the graphics multiple buffer render buffer into a plurality of buffers, wherein the buffers of the graphics multiple buffer render buffer are based on the number of frame buffers identified in the buffer request; and determine a number of display devices connected to the information handling system; in response to the number of display devices being below a threshold number of display devices, create a plurality of frame buffers, wherein a number of the frame buffers is the same as a number of the buffers of the graphics multiple buffer render buffer, wherein each of the frame buffers provides data to a different one of the first, second, third, fourth, fifth, and sixth display ports; and in response to the number of display devices being above threshold number of display devices, perform a buffer management logic prior to the plurality of frame buffers being created.

18. The information handling system of claim 17, wherein the processor further to: detect an output resolution for each of the of the first, second, third, fourth, fifth, and sixth display ports.

19. The information handling system of claim 17, wherein the frame buffers are physical buffers within the information handling system.

20. The information handling system of claim 17, wherein each of the first, second, third, fourth, fifth, and sixth display ports outputs 4k resolution data.
